

PROCESS ENGINEER

PERSONAL INFORMATION

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KEY SKILLS

Technical

- Preparation and review of the following technical documents: Process Drawings PFDs/ MSDs/ P&IDs, Heat and Material Balance, Process Description, Calculation notes, simulations.
- Main equipment sizing such as heat exchangers, pumps, drums, and equipment list establishment.
- Process specifications preparation.
- Process Hydraulic profile achievement.
- Line, valves & instrumentation sizing, and associated lists establishment.
- Process simulation under AFT arrow, Aspen Hysys/Plus, Aspen Heat Exchanger Design and Rating & PROII.
- HAZOP reviews.

IT

Process software: Aspen Heat Exchanger Design and Rating, Aspen Hysys/Plus, Pro II, ProSim, Aveva Diagrams, Aveva Engineering, Smart Sketch, EtaPRO, PRISM, PI ProcessBook, Navisworks Freedom, AFT Arrow, DepoCal, AutoCAD, AutoCAD Plant 3D.

Others: Fortran, Word, Excel/VBA, Outlook, PowerPoint, Visio.

Management

- Manage and coordinate the execution of the project.
- Working in direct interface with the client (Manager, engineers, experts & Operations and Maintenance teams).
- Perform detailed technical clarifications and establish key project documents.
- Project implementation schedule and budget.
- Definition of the relevant technical solutions,
- Freezing of the project basis of design in compliance with the Management System for all technical, quality and safety aspects.
- Establishing project final scope of work, risk assessment, budget and time schedule.
- Implementation of new quality management system certification for the nuclear sector.

PROFESSIONAL EXPERIENCES

JACOBS (June 2020 – ongoing)

Process Engineer / site coordinator – EDF – France

June 2020 – ongoing

Project context

Detailed Design – Review of System Protection Analysis SPA for EPR2 fluid systems to ensure their mechanical robustness against the risk of exceeding admissible limits (pressure and temperature) based on the essential requirements of ESPN order with internal standards and guidelines.

Role and Responsibilities

- Task order technical-financial offer preparation and presentation to the client and site coordination.
- SPA review comments and questions deliverable:
 - Comments and questions list preparation for the studied system SPA based on the following inputs: SPA, loading specification, design report, process diagrams, P&IDs & architecture description including CODARR meeting outputs.
- Presentation of relevant questions and comments during CODARR meeting.
- Definition of the relevant technical solutions.

Project context

DMS (Disruption Mitigation System for Pellet Generation Process) Project Design study:

- Manifold dimensioning calculations.
- Parametric calculator creation to determine the overall duration of the Gas Handling Process Timing of DMS readiness following a plasma pulse.
- Modelling of pellet gas and pellet shell gas flow to the cold head for pellet formation.

Role and Responsibilities

- Production and reviewing the following deliverables:
 - Technical specification.
 - Input data notes.
 - Native calculation files (AFT Arrow models and Excel files).
 - Technical Memo presenting model setup under AFT Arrow, brief description of each case to study and the results based on AFT arrow modelling outputs with recommendations.
 - Technical Memo presenting the calculator construction and user instructions.
 - Technical memo with summary of inputs, assumptions, methodology, results with key outputs, and any engineering recommendations relating to process control or process sizing.
 - freezing of the project basis of design in compliance with the Management System for all technical, quality and safety aspects.
 - Progress meetings slides and Minutes of Meeting.

Project context

Design, Procurement, Installation, Test and Commissioning of the Radiological Environmental Monitoring System (REMS) - ITER First Plasma Stage

Role and Responsibilities

- Preparation of the following deliverables:
 - PFDs and P&IDs process diagrams for gas sampling systems (stack/ventilation duct & room ambient air).
 - Quality assurance and equipment technical specifications.
 - Calculating pressure drop and aerosol particle deposition methodology definition.
 - Requirements compliance matrix.
 - List of codes, standards, and regulations.
 - Input data completeness assessment report.
- Review of the 3D model of the air sampling systems.
- Sizing of equipment and lines.
- Calculation of aerosol particle deposition rate in the upstream line of each sampler.
- Equipment and valve lists.
- Procurement activities initiation.

Project context

Basic Design – WWT wastewater treatment project

Role and Responsibilities

- Elaboration of the following deliverables:
 - PI&Ds process diagrams for utilities: chemicals, compressed air, drinking water, industrial water.
 - Preliminary routings for the following networks: steam, condensates, industrial water, drinking water, hot water, glycol water & cooling water.
 - Process description note.
 - Equipment & line sizing.
 - Equipment, lines, valves, and instrumentation lists.

Project context

Basic Design – Review of System Protection Analysis SPA for EPR2 fluid systems to ensure their mechanical robustness against the risk of exceeding admissible limits (pressure and temperature) based on the essential requirements of ESPN order with internal standards and guidelines.

Role and Responsibilities

- SPA review comments and questions deliverable:
 - Comments and questions list preparation for the studied system SPA based on the following inputs: SPA, loading specification, design report, process diagrams, P&IDs & architecture description including CODARR meeting outputs.
 - Summary of:
 - o pending actions to be done on the studied system and on the interfacing systems.
 - o residual risks identified in studied system SPA.
 - o existing and new safety accessories protecting the studied system or the interfaced system against pressure and temperature overpassing.
- Presentation of relevant questions and comments during CODARR meeting.

Vulcain Engineering (July 2019 - March 2020)

Process Engineer – TECHNIP Energies – France

October 2019 – January 2020

Project context

FEED – Amur Gas Cracker Complex (AGCC), Mixed Gas steam cracking process design for ethylene/ Propylene production - SIBUR (Russia)

Role and Responsibilities

- Equipment sizing and calculation notes & equipment process datasheets preparation (filters, mixers, heat exchangers, drums) and equipment list update.
- Line sizing for the hot section.
- Process drawings mark-ups (P&IDs, PFDs, MSDs).
- HAZard and OPerability Study (HAZOP) for cracking unit hot section.

Process Engineer – AXENS – France

July 2019 – October 2019

Project context

Basic Design – Complex crude Oil To Chemicals – Saudi Aramco/ SABIC.

Role and Responsibilities

- Atmospheric distillation and vacuum distillation units:
 - Lines sizing and lines list preparation,
 - Instruments list preparation,
 - Hydraulic profiles production
 - Equipment Process datasheets preparation,
 - Material Selection Diagram update and Material Selection Guide preparation.
- Gas plant unit:
 - Hydraulic profile review.

Capgemini Engineering (July 2016 – July 2019)

Process Engineer/ site coordinator – EDF INGEUM – France

June 2017 – July 2019

Project context

E-monitoring – Thermal Power Plants (Gas turbine and Combined gas).

Role and Responsibilities

- Combined gas cycle and gas turbines key performance indicators calculation, analysis, and monitoring.
- Writing plants performance reports providing sites with technical feedback regarding equipment/plant performance & degradation with associated recommendations.
- Long-term performance analysis for combined gas cycle and gas turbines.
- Carrying out in-depth studies on performance losses, proposing solutions to improve performance in collaboration with EDF Group experts.
- Early fault detection weekly for the monitored combined gas cycles and gas turbines.
- Monthly exchanges by conference-calls with performance site engineers and operators on any issues encountered and associated recommendations.
- Site coordination

Project context

Basic Design – Design of new radioactive waste treatment plant for recycling nuclear materials (La Hague).

Role and Responsibilities

- Preparation and review of several technical documents for chemicals & utilities:
 - Process drawing (PFDs & P&IDs).
 - Process description documents.
 - Process Information Sheets for Instrumentation (sensors, control, fluid characteristic).
 - Process Information documents for automatism (main operations/ equipment).
 - Supplier contact.

IPS Belgium (Apr 2016 – June 2016)**Project context**

Feasibility - Design of a new molybdenum purification unit.

Role and Responsibilities

- Heat and Material Balance.
- Preliminary Process Flow Diagram preparation.
- Equipment (reactors, exchangers, pumps) sizing.
- Process Description.

TotalEnergies (July 2013 – January 2014)**Project context**

Crude distillation unit performance monitoring, optimization, and technical support.

Role and Responsibilities

- Crude Oil desalting and distillation operations key performance indicators monitoring in real time.
- Optimization of distillation column top temperature.
- Process simulation under Pro II software.
- Writing distillation unit performance monthly reports.
- Vacuum Distillation Unit furnace decoking operation follow up.
- Fluid Catalytic Cracking unit main fractionator revamp (trays & internals) follow up.
- Crude Distillation Unit and Diesel Hydrotreating Unit HAZOP reviews.

EDUCATION

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| 2014 | master's degree – Chemical Engineering – Conservatoire National des Arts et Métiers – France |
| 2011 | master's degree - Pharmaceutical Chemistry – Paris South 11 University – France |
| 2009 | engineering degree – Process Engineering – M'Hamed Bougara University – Algeria |

TRAINING

PRO II simulation - IFP training – TotalEnergies Refinery (Antwerp- Belgium)
English training - Wall Street English (Paris - France)
Aveva Diagrams & Aveva Engineering – Orano/Jacobs (Paris - France)
Introduction to Project Management - Orsys (Paris - France)

LANGUAGES

English: professional

Arabic: advanced

French/ Kabyle: native

HOBBIES

Sport, cinema, reading, travelling, music